WPRS — Whisper Position Reporting System

Position beaconing over HF with a 200 mW WSPR transmitter on a car: a two-page summary

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The idea. A WSPR transmitter that computes its Maidenhead locator from a GPS fix turns the worldwide network of volunteer WSPR receivers into a free position channel: no mobile network, no APRS digipeater and no receiver of one's own is needed. Somebody, somewhere, hears the transmission and uploads it to the public WSPR database, and the database is the track. This summary describes the technique and what it achieved from an ordinary car in three summer seasons: 42 723 logged transmissions from 2024-07-04 to 2026-11-06.

How it works

A WSPR transmission lasts 110.6 s, occupies about 6 Hz in the 200 Hz WSPR segment (14 097.0–14 097.2 kHz on 20 m) and is decoded down to about −30 dB SNR in 2 500 Hz. The message carries 50 bits; a *Type 3* message holds a six-character locator, a hash of the callsign and the power. A six-character locator (a *subsquare*, for example KP20jg) is 4.6×4.6 km at 60°N, so the position is known with an r.m.s. error of 1.9 km (at most 3.3 km). The four-character locator of an ordinary message would be 111 km across. A fix reaches the map via the receiver, the WSPR database and `wspr.live`; the WSPR Logger polls every ten minutes, so a position is visible about 6–10 minutes after it was sent. WPRS is a *breadcrumb trail*, not a live tracker.

Station

Transmitter: ZachTek WSPR desktop transmitter, about 200 mW (23 dBm), built-in GPS with an external active antenna, one transmission every ten minutes on 20 m. **Antenna:** Diamond monoband 14 MHz whip on a magnetic mount on the roof of a VW Touran (the car body is the ground plane; not fitted in winter, so the data cover March to November). **Power:** a 30 000 mAh USB power bank. At the measured duty cycle (18.4 % transmitting) the average load is 0.64 W and the predicted runtime 6.2 days; in practice a pack lasts four to five days, and two packs are rotated.

Results

Table 1. Key measurements, three seasons (2024–2026). A *fix* is a transmission heard by at least one station.

Data	42 723 fixes on 490 of 856 days; 102 different six-character locators; 3 713 receiving stations
Cadence	median 91 fixes per day (63 % of the 144 possible); 91 % of consecutive fixes exactly 10 min apart
Availability	a fix from the previous 60 min: 92 % of the time at 08–16 UTC, 38 % at 01–04 UTC (lowest 31 % at 2 UTC)
Confirmation	median 17 receivers per fix; 85 % heard by three or more stations, 9 % by a single one
Position	home square KP20jg 86 %, adjacent squares 10 %, more than 25 km from home 2 %; farthest fix 193 km (KP31kq, 2026-06-19); 11 trips beyond 25 km
Coverage	83.5 % of the receivers in Europe; a station beyond 6 000 km on 73 % of days; record 15 619 km (78 095 km per watt)
Lower power	stations still decoding (upper bound): 50 mW 98 %, 25 mW 89 %, 10 mW 70 %, 1 mW 27 %
Seasons	fixes per median day 81, 90, 96; heard by ≥3 stations 81, 86, 86 % (2024, 2025, 2026)

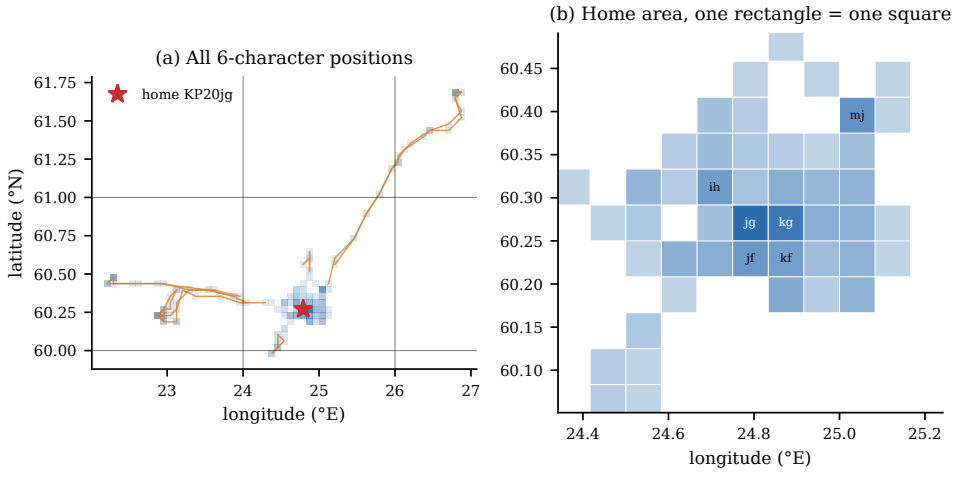


Figure 1. (a) Every six-character position recorded, shaded by number of fixes, with the longest trips. (b) The home area; each rectangle is one subsquare of 4.6×4.6 km.

Availability follows the band. The transmitter sends every ten minutes; a fix exists when at least one station decodes it. In the daytime the latest position is almost always less than an hour old; in the small hours, when 20 m is closed, it often is not (Figure 2). Fixes are well confirmed: a typical one is heard by 17 independent receivers, which makes a wrong position from a single bad report rare.

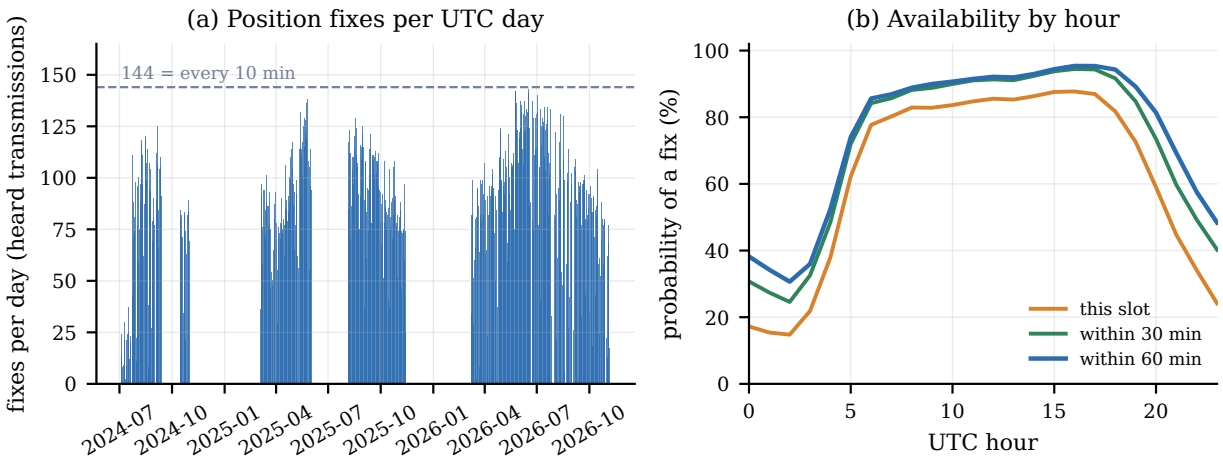


Figure 2. (a) Position fixes per UTC day over the three seasons. (b) Probability of a fix by UTC hour: from the current slot, from the previous 30 and the previous 60 minutes.

The track is real. Between 06 and 14 UTC the car is in the home square in 67 % of the fixes on weekdays but 88 % at weekends: the commute is recovered from the fixes alone. Trips reached 193 km from home. Two places closer than about five kilometres cannot be told apart; that is the resolution limit of the technique. Most fixes are delivered by European stations one to three thousand kilometres away; the long-distance reports prove the reach but are not needed for tracking. Because European reception is so redundant, even 10 mW would still have reached 70 % of the stations; the long paths would be lost first.

Where WPRS fits

Compared with APRS (needs a VHF digipeater or gateway in range) and a cellular GPS tracker (needs mobile coverage and a subscription), WPRS needs no infrastructure and only 0.2 W, but delivers a square of about 5 km every ten minutes, with no reverse channel and a gap at night. It suits *away-from-coverage breadcrumb trails* (long drives, hiking, sailing, expeditions) and doubles as a propagation probe. It is not a safety tracker and cannot replace a distress beacon.

Limitations. One car, one antenna and three seasons; the two earlier seasons were rebuilt from `wspr.live` (checked against a live week: ratio 1.001). No ground-truth GPS log was kept, so the position error is the theoretical one of the locator. A missing fix can mean a closed band, a lost GPS fix, a battery swap or a garage. The track is public: anybody can see where the car is to within about five kilometres, so switch the transmitter off at home if that matters.

Operating tips. Wait for the GPS lock before driving off; give the GPS puck a clear view of the sky; keep the magnetic mount on clean roof metal; do not rely on the latest fix at night; carry a second charged power bank on trips longer than four days.

In short: a 200 mW GPS WSPR transmitter with a car-roof whip is a usable infrastructure-free position beacon on 20 m. It gives a 4.6 km square every ten minutes, a position from the last hour 92 % of the daytime and 38 % of the night, confirmed by a median of 17 receivers, and runs four to five days on a 30 000 mAh power bank.